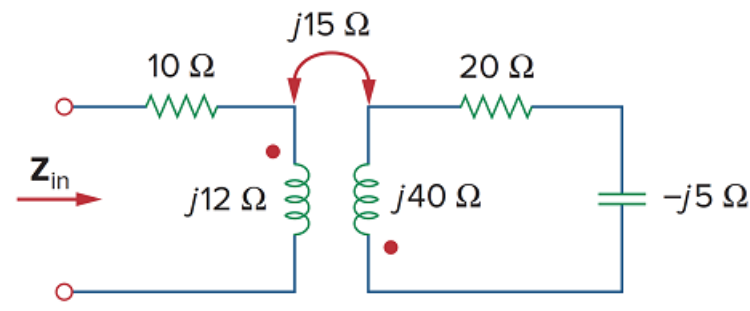


Problem

Determine the input impedance of the air-core transformer circuit of Fig. 13.102.

Figure 13.102



Step-by-step solution

Step 1 of 2

Refer to Figure 13.102 in the textbook.

Refer to Figure 13.19 for a linear transformer circuit which is similar to Figure 13.102.

Refer to equation 13.41 from the textbook for input impedance formula.

Write the expression for input impedance of the transformer circuit.

$$\mathbf{Z}_{\text{in}} = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L} \dots\dots (1)$$

Here,

R_1 is the resistance of primary winding

$j\omega L_1$ is the inductive reactance of primary winding

R_2 is the resistance of secondary winding

$j\omega L_2$ is the inductive reactance of secondary winding

$j\omega M$ is the mutual inductive reactance

$\mathbf{Z}_L$ is the load impedance

Step 2 of 2

Substitute $10\ \Omega$ for R_1 , $j12\ \Omega$ for $j\omega L_1$, $15\ \Omega$ for M , $20\ \Omega$ for R_2 , $j40\ \Omega$ for $j\omega L_2$ and $-j5\ \Omega$ for $\mathbf{Z}_L$ in equation (1).

$$\begin{aligned} \mathbf{Z}_{\text{in}} &= 10 + j12 + \frac{(15)^2}{20 + j40 - j5} \\ &= 10 + j12 + \frac{225}{20 + j35} \\ &= 10 + j12 + 2.769 - j4.846 \\ &= 12.769 + j7.154\ \Omega \end{aligned}$$

Therefore, the input impedance of the air-core transformer circuit, $\mathbf{Z}_{\text{in}}$ is $12.769 + j7.154\ \Omega$.